

Linglin He

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About Me

I am a third-year undergraduate student in Computer Science at SUSTech. My research focuses on multimodal information retrieval and efficient vision-language systems, particularly visual document retrieval and representation compression, under the supervision of [Prof. Xuemeng Song](#). I am also interested in exploring LLM agents, multimodal reasoning, and spatial intelligence.

Education

Southern University of Science and Technology (SUSTech)

Sep 2024 – Present

BS in Computer Science

GPA: 3.81/4.0, Member of the Turing Class (Top 1% of the CS department, designated for elite CS students at SUSTech).

Professional Skills

- ML & Programming: Python, C++, PyTorch, JAX/MaxText, Multimodal Retrieval, Representation Compression
- Systems: Linux/Shell, MPI, SLURM, Distributed Training (FSDP), Docker, FastAPI/LangGraph

Publication

1. **Linglin He**, Haokun Wen, Haocheng Dou, Na Zheng, Linyi Yang, Xuemei Song. *RetVDQ: Retrieval-Aware Variable-Depth Residual Quantization for Visual Document Retrieval*. Under review at ICLR 2027.
2. Fulin Wu, **Linglin He**, Shulin He, Zhongqiu Wang. *A Microphone Padding Approach for Speech Separation with Varying Number of Microphones*. NCMMSC 2026, Oral Presentation.

Research & Engineering Projects

RetVDQ: Retrieval-Aware Variable-Depth Residual Quantization for Visual Document Retrieval

Undergraduate Researcher, Advisor: [Prof. Xuemeng Song](#)

Jun 2026 – Sep 2026

First-author paper under review at ICLR 2027

- Proposed RetVDQ, a retrieval-aware variable-depth residual quantization framework that adaptively allocates precision across document embeddings while preserving retrieval rankings.
- Retained 99.3% of full-precision retrieval quality with only 3.2% of the original storage on average across ViDoRe V1 and V2.

Autonomous AI Student Productivity Agent

Team Course Project

Feb 2026 – Jun 2026

Awarded full marks including all bonus points

- Built the core autonomous AI agent system from scratch with FastAPI and LangGraph, implementing stateful agent execution, long-term memory, context management, tool use, and human-in-the-loop control.
- Developed and optimized agent workflows for scheduling and information retrieval, including conflict resolution, intent-aware routing, selective tool loading, and execution reliability mechanisms.

A Microphone Padding Approach for Speech Separation with Varying Number of Microphones

Core Member, Advisor: [Prof. Zhong-Qiu Wang](#)

Jan 2026 – Mar 2026

- Implemented and experimentally validated a microphone-padding framework for TF-GridNet, contributing to method refinement and conducting extensive benchmarking and ablation studies across varying microphone configurations.

ISC 2026 Student Cluster Competition

SUSTech Supercomputing Team, Sole Contributor(AI Track)

Mar 2026 – Jun 2026

Team received the Achievement Award

- Solely completed AI Track, optimizing distributed continued pre-training of Qwen3-14B with JAX/MaxText on 4× NVIDIA GH200 GPUs.
- Tuned and profiled FSDP-based training across parallelism, rematerialization, batch size, and communication, achieving 9.89K tokens/s and identifying key communication–computation bottlenecks.

Performance Optimization for CloverLeaf CFD Application

HPC Competition Project, Solely Contributor (CloverLeaf Track)

Jul 2025 – Aug 2025

- Solely optimized the CloverLeaf CFD workload across kernel, compiler, MPI, and NUMA levels, achieving up to 4.19× speedup and reducing large-scale multi-node runtime to 370.04s.

Robust Audio Deepfake Detection — ICME 2026 ESDD2 Challenge

Undergraduate Researcher, Advisor: [Prof. Zhong-Qiu Wang](#)

Feb 2026 – Jun 2026

- Implemented and optimized a robust audio deepfake detection pipeline, resolving collapsed XLSR feature fusion and reducing environmental-spoof EER from 0.44 to 0.07 through extensive training and optimizer refinement.

Internship Experience

Software Engineering Intern, Liii STEM (MoganLab)

Aug 2025

- Implemented and optimized core library functions for the Goldfish Scheme R7RS interpreter, contributing multiple peer-reviewed pull requests to the main branch while maintaining full test and standards compliance.

Extracurricular Activities

SUSTech Supercomputing Team, Member

2026 – Present

- Represented SUSTech in the ISC 2026 Student Cluster Competition and is participating in the APAC HPC-AI Competition.

Selected Awards

- First Prize, Guangdong, 2025 Undergraduate Contest in Mathematical Modeling
- First-class Excellent Student Scholarship, 2024-2025 Academic Year (Top 3.6%)
- Achievement Award, ISC 2026 Student Cluster Competition
- Runner-up, Fundamental Track, 2nd SUSTech Supercomputing Competition
- Outstanding Youth League Leader, 2024 Academic Year
- Outstanding Youth League Leader, 2025 Academic Year

English Proficiency

TOEFL iBT (Feb 2026): 5.0/6.0